

Lecture #2

Last modified: September 15, 1998

Abstract

0.1. Old Picture

1. Modelling randomness

1. What is random-ness

1. roulette wheel type randomness- an event about which we have no idea - it is not possible for anyone to know it

1. roulette wheel
2. horse race
3. football game
4. dumb luck

2. something inside a box

1. when the phone rings
2. get a letter from a college on April 15
3. auto mechanic

3. Drawing

2. The state-space model

States of the world

Definition 1. *An experiment is a situation where one of several possible outcomes is revealed in the future.*

Remark 1. *Don't think narrowly about experiments as lab experiments. It could be something that is revealed in the future but could in principle be known now.*

Example 1. *Will it rain or not rain tomorrow? or Is it raining right now?*

Example 2. *Will the economy grow, shrink, or stay the same next quarter? Or is it growing, shrinking or staying the same right now?*

Example 3. *Yankees win, Yankees lose*

Example 4. *My car is making a funny noise. There is something wrong with it or there is nothing wrong with it.*

Example 5. *General Motors stock goes up, stays the same, or falls over the next year. GM stock goes up or down, while I am talking to you. It was at 68 when I left my office.*

Example 6. *Three horses, Horse A, Horse B and Horse C*

Order	1	2	3	4	5	6
First place	A	A	B	B	C	C
Second place	B	C	A	C	A	B
Third place	C	B	C	A	B	A

6 possible outcomes (assuming that all the horses finish and no ties are allowed).

Definition 2. *We call each outcome a state of the world.*

Definition 3. *The set of all possible outcomes of an experiment is called the sample space.*

1. Any possible outcome must be in some state.

Example 7. *it can either rain or not rain but it cannot do neither.*

2. Mutual exclusivity No outcome can be in more than one state.

Example 8. *it can't both rain and not rain*

Example 9. *Suppose we toss two coins a nickel and a dime.*

State	Nickel	Dime
s_1	Heads	Heads
s_2	Heads	Tails
s_3	Tails	Heads
s_4	Tails	Tails

3. Probability measures

1. What can we say about these outcomes?
 1. The economy will probably grow
 2. My car probably has a problem
 3. The Yankees will almost surely get the American League wild card
 4. The Tigers are definitely not going
 5. "I think it's a fifty-fifty shot"
2. The notion of probability
 1. assign a number to a state
 2. numbers are between zero and one
 3. numbers add up to one
 4. things that are more likely have higher number
3. Do not think too much about this, assume that probability is handed down from god.

Definition 4. *A probability measure is a function that assigns a probability to each state.*

$$p(s)$$

1. It is between zero and one (or 100%)

$$0 \leq p(s) \leq 1$$

2. The sum of the probability of every state is one

$$\sum_{s=1}^S p(s) = 1$$

3. States that never happen have zero probability
4. A state that is more likely has a higher probability:

$$\text{state 1 is more likely than state 2} \iff p(s_1) > p(s_2)$$

Example 10. Suppose we think about our coin tosses again.

<i>State</i>	<i>Nickel</i>	<i>Dime</i>
s_1	<i>Heads</i>	<i>Heads</i>
s_2	<i>Heads</i>	<i>Tails</i>
s_3	<i>Tails</i>	<i>Heads</i>
s_4	<i>Tails</i>	<i>Tails</i>

1. A probability measure for this situation could be: $p(s_i) = 0.25$ for all i .
2. Note that the probability of each event is greater than zero and less than one,
3. The sum of the probability of the states is one.
4. Now suppose the dime has tails on both sides.
5. Then this would not be a probability measure for this situation, because s_1 and s_3 cannot happen, so their probability should be zero.

4. Events

Definition 5. An event is a subset of the set of states.

Example 11. Consider the coin-toss world again:

<i>State</i>	<i>Nickel</i>	<i>Dime</i>
s_1	<i>Heads</i>	<i>Heads</i>
s_2	<i>Heads</i>	<i>Tails</i>
s_3	<i>Tails</i>	<i>Heads</i>
s_4	<i>Tails</i>	<i>Tails</i>

The set of events is $\{s_1, s_2, s_3, s_4\}$. Some possible events are:

1. Nickel turns up heads $E_1 = \{s_1, s_2\}$
2. Dime turns up heads $E_2 = \{s_1, s_3\}$
3. At least one coin turns up tails $E_3 = \{s_2, s_3, s_4\}$
4. At least one coin turns up heads $E_4 = \{s_1, s_2, s_3\}$

Now we will do a little set theory (very little).

4.1. Union of two events

Definition 6. An event E is the union of two events A and B denoted $A \cup B$ if it contains all the states in both A and B . I.e.

$$s_i \in A \cup B \Leftrightarrow s_i \in A \text{ and } / \text{ or } s_i \in B$$

Example 12. From our coin-toss world above, event $E_4 = \{s_1, s_2, s_3\}$ is the union of E_1 and E_2 because every member of both sets is contained in it.

4.2. Intersection of two events

Definition 7. An event E is the intersection of two events A and B denoted $A \cap B$ or AB if it contains only states which are in both A and B .

$$s_i \in A \cap B \Leftrightarrow s_i \in A \text{ and } s_i \in B$$

Example 13. event $E_5 = \{s_2\} = E_1 \cap E_3$

4.3. Probability of an event

Definition 8. the probability of an event is simply the sum of the probabilities of the individual states which it comprises. Let's say that event E is the states 1 and 2.

$$p(E) = \sum_{i=1}^2 p(s_i)$$

1. Events are not mutually exclusive. The probability of two events is not necessarily the sum of the probabilities of the two events. The probability of two events is the probability of their union.

Example 14. Consider our coin-toss problem again. $E_4 = \{s_1, s_2, s_3\}$ and $E_3 = \{s_2, s_3, s_4\}$. So the probability of E_4 is

$$p(E_4) = \sum_{i=1}^3 p(s_i) = p(s_1) + p(s_2) + p(s_3) = .75$$

The probability of E_3 is:

$$p(E_3) = \sum_{i=2}^4 p(s_i) = p(s_2) + p(s_3) + p(s_4) = .75$$

Proposition 1. The probability of an event $E = A \cup B$ is the probability of one event plus the probability of the other event minus the probability of both.

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

This implies that

$$p(A \cap B) = p(A) + p(B) - p(A \cup B)$$

Example 15. 100 workers. We know that some smoke and some are absent a lot. We choose one at random. The four possible outcomes are:

Number of days absent	smoker	non-smoker
Less than 10	s_1	s_2
10 or more	s_3	s_4

1. Experiment is choosing a person
2. Some events are $E_1 = \{\text{smokers}\} = s_1 \cup s_3$, $E_2 = \{\text{non-smokers}\} = s_2 \cup s_4$,
 $E_3 = \{< 10\} = s_1 \cup s_2$, $E_4 = \{\geq 10\} = s_3 \cup s_4$,
3. Assign probability $p(s_1) = \frac{8}{100}$, $p(s_2) = \frac{20}{100}$, $p(s_3) = \frac{65}{100}$, $p(s_4) = \frac{7}{100}$
4. Then, $p(E_1) = p(s_1) + p(s_3) - \underbrace{p(s_1 \cap s_3)}_{=0} = p(s_1) + p(s_3)$

4.4. Conditional probability and independence

In some situations we might know that one event will occur. We might want to know the probability that another event will occur given that that event has already occurred.

Example 16. Suppose we knew that the nickel would turn up heads, is the dime now more or less likely to turn up heads.

Example 17. GM stock went up. Is Ford stock more or less likely than we previously thought to have gone up?

Example 18. In the example above, if we choose a smoker, is he more or less likely to have been absent more than ten days than he would have been otherwise.

We call this conditional probability. We can understand it through a Venn-Diagram

Formally,

Proposition 2. The conditional probability that X will occur if Y has already occurred, denoted $P(X | Y)$ is

$$P(X | Y) = \frac{P(X \cap Y)}{P(Y)}$$

Example 19. What is the probability that we will choose someone who has less than 10 sick days if we know that he/she is a smoker.

$$P(E_3 | E_1) = \frac{P(E_3 \cap E_1)}{P(E_1)}$$

Since $E_3 \cap E_1 = s_1$, and $E_1 = s_1 \cup s_3$, this is

$$\frac{p(s_1)}{p(s_1 \cup s_3)} = \frac{\frac{8}{100}}{\frac{73}{100}} = \frac{8}{73}$$

Definition 9. We say that two events are independent if the probability of an event happening is the same conditional on the event as it is unconditionally

$$p(X | Y) = p(X)$$

Since

$$P(X | Y) = \frac{P(X \cap Y)}{P(Y)}$$

under independence

$$p(X) = \frac{P(X \cap Y)}{P(Y)}$$

Which implies that

$$P(X \cap Y) = p(X)p(Y)$$

The probability of both events happening is the product of the probability of the two events.

Example 20. In the preceding example, the two events are not independent

Example 21. (The gambler's fallacy) Consider the coin-toss world again:

State	Nickel	Dime
s_1	Heads	Heads
s_2	Heads	Tails
s_3	Tails	Heads
s_4	Tails	Tails

Assume $p(s_i) = .25$. Event 1 is Nickel turns up heads $E_1 = \{s_1, s_2\}$. Event 2 is Dime turns up heads $E_2 = \{s_1, s_3\}$. What is the probability of a nickel turning up heads if we know that the dime is heads.

$$p(E_1 | E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{p(s_1)}{p(s_1) + p(s_3)} = \frac{.25}{.5} = .5$$

Note that these events are independent because $p(E_1) = .5 = p(E_1 | E_2)$. Note also that $P(E_1 \cap E_2) = .25 = P(E_1)P(E_2)$